AI Lab Documentation

Generative AI Practice and StepWise, a Design-Thinking Math Tutor Bot

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AIML-500: Machine Learning Fundamentals, Assignment 1.4

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Link to product (Mizou chatbot): https://mizou.com/login-thread?ID=y9RHjnmmG4M-q49uMddTqN06Q9Yqo_NTN8sAH_jD-fo-153879

1. Overview

This lab had two parts. First, I practiced with three kinds of generative AI tools (a general LLM, a custom GPT, and a research assistant) and compared their outputs. Second, I used design thinking to plan, build, test, and iterate on my own AI chatbot. I chose a problem from my own life: I tutor math, and my students often get stuck between sessions on setting up word problems. The result is StepWise, a Socratic tutor that gives hints instead of answers.

Activity	Tool used	Purpose
Practice with an LLM	ChatGPT (free, logged out)	Same Context/Task/Constraints prompt in 3 separate chats
Practice with a custom GPT	Tutor Me by Khan Academy (GPT store)	Change, add opinion, ask for more, ask why, what-if, summarize
Research assistant	STORM (Stanford OVAL)	Generate a cited article on a topic I know well; review its brainstorming
Build a bot	Mizou (free)	StepWise: Word Problem Coach, built and tuned over 4 versions

2. Practice with an LLM

Prompt (Context, Task, Constraints)

```
Context: I tutor high school and college students in math. Many of them get stuck on
word problems because they don't know how to turn the words into an equation.
Task: Explain a simple 4-step method for setting up a linear equation from a word
problem, then walk through one worked example about two friends splitting a cost.
Constraints: Keep it under 250 words, use plain language a 10th grader can follow, and
end with one practice problem without giving the answer.
```

I ran this exact prompt in three separate, fresh ChatGPT chats so no earlier conversation could influence the results.

Comparison of the three outputs

	Chat 1	Chat 2	Chat 3
4 steps	Unknown, Info, Translate (keyword list), Write equation	Unknown, Info, Write equation, Check equation	Unknown, Info, Choose operation, Write + check
Worked example	Equal split of $60: x + x = 60, x = 30	$50 bill, one pays $10 more: x + (x + 10) = 50	$48 bill, one pays $6 more; solved to $21 and $27
Solved example?	Yes, but trivial	No, stopped at the equation	Yes, fully
Practice problem	Equal split of $36 pizza (nearly identical)	$72 game, older pays $12 more	$60 bill, one pays $10 more, with a hint
Style	Headers and keyword list	Clear; ended with a login prompt	Most complete; added an emoji

Were the results what I expected?

Mostly. All three followed the structure, the 10th-grade reading level, and the no-answer practice problem. I didn't expect how much the actual content changed: each chat invented a different fourth step, and only Chat 3 gave a worked example that was both realistic and fully solved.

Significant differences and similarities

Similar: same opening steps (find the unknown, find the information), the same friendly tone, and a practice problem that mirrored the example. Different: the method itself (keyword translation vs. checking vs. choosing an operation), the difficulty of the example (an equal split is too easy to need a method), and completeness (Chat 2 never solved its example). The same prompt produced three different lessons, so a single output can't be treated as "the" answer.

Criteria I used to judge quality

(1) Accuracy of the math. (2) Teaching value: does the example actually need the method? (3) Completeness: was the example solved and checked? (4) Constraint-following: length, reading level, practice problem without an answer. (5) Usability for a real student. By these criteria Chat 3 was best, Chat 2 second (good example, incomplete), and Chat 1 weakest (too easy, practice nearly copied the example).

Reflection

LLM output is a sample, not a fixed result. Running the same prompt several times and comparing is a cheap way to find the best version and to spot what the model is unsure about. It also showed me that keyword lists ("total means add") are a common default, which I think can mislead students. That became a design decision for my bot.

3. Practice with a Custom GPT

I chose Tutor Me, Khan Academy's free GPT (rated 4.5, #15 in Education, 3M+ conversations), because it is the closest existing product to the bot I planned to build. A banner in the GPT store noted that GPTs will be retired on December 11 and moved to "plugins," a reminder of how fast these platforms change.

Required move	What I did	What happened
Conversation	Gave the $50 bill / $10-more problem as a student and answered its questions	It never gave the answer. It asked one question at a time and confirmed my correct setup x + (x + 10) = 50.
Ask it to change something	Switch to a $72 video game where the older sibling pays $12 more	Rewrote the problem and let me try it on my own.
Add my own opinion	Said that, as a tutor, I think keyword lists confuse students because "less than" flips the order	Strongly agreed, suggested picturing the story or drawing bars, then started asking me questions as a fellow tutor.
Ask for more / better	Asked for a better way to check answers than plugging back in	Suggested "tell the story back" and the phrase "Right people. Right difference. Right total."
Ask why	Why did it ask about each share before any equation?	Build meaning first, give the variable a purpose, let the student build the equation (mathematical modeling).
Ask "what if...?"	Three friends, one pays double the other two	Set up x, x, 2x, but did not notice my problem had no total, so it can't be solved.
Summarize	Asked for a summary	Accurate 7-bullet summary ending with

Required move	What I did	What happened
		"understand the story first, build the math second."

Reflection

Tutor Me showed what a well-instructed GPT can do: consistent Socratic questioning, a warm tone, and adapting to who it's talking to. It also showed weaknesses I wanted to avoid: it agreed with my opinion without pushing back, it missed that my what-if problem was unsolvable, and the free version showed an ad mid-conversation. I borrowed the "right people, right difference, right total" check for my bot and added a rule to check whether a problem has enough information before starting.

4. Practice with a Research Assistant (STORM)

I picked a topic I know from experience so I could evaluate the output: retail demand forecasting with ARIMA, LSTM, and XGBoost. I built forecasting prototypes with these models at a warehouse and order-management startup. STORM required a Google sign-in and acceptance of its terms, which say inputs may be used for research and shared under a CC-BY license, so I avoided entering any personal information.

Process

After I entered the topic and a writing purpose (compare the three approaches: when each works, data needs, tradeoffs), STORM showed four stages over about three minutes: (1) identifying perspectives, (2) browsing the internet, (3) organizing information into a hierarchical outline, and (4) writing the article. The result was a Wikipedia-style article with a table of contents (methods, workflow, evaluation, applications, comparison, limitations) and 26 numbered citations.

My evaluation of the article

- **Strengths:** Accurate on ARIMA/SARIMA (p, d, q; differencing; Ljung-Box residual checks). Strong on what practitioners actually worry about: time-ordered (walk-forward) validation, data leakage from overlapping windows or fitting scalers on all the data, MAPE breaking down near zero demand, intermittent demand, and the point that lower forecast error does not automatically mean better inventory decisions. This matches my own experience.
- **Weaknesses:** The LSTM section was very thin, saying its sources didn't specify an architecture. There was no clear practical recommendation for when to pick each model, some repetition across sections, and nothing on newer options such as hybrid models or time-series foundation models.

The brainstorming process

The "See BrainSTORMing Process" view showed four simulated personas: a machine-learning practitioner, a basic fact writer, a time-series statistician, and a deep-learning researcher. Each persona asked questions, and a simulated expert answered using only retrieved sources, with citations. The expert openly said when the sources didn't cover something (for example, how XGBoost encodes calendar and promotion features). Interestingly, the deep-learning persona

did gather details on LSTM gates and memory, but little of it reached the final article, so not everything gathered in brainstorming made it through.

How I think STORM works

In my own words: an LLM first generates several perspectives on the topic. For each perspective, two AI agents hold a simulated interview, a "writer" asking questions and an "expert" that runs web searches and answers only from what it finds. All the questions, answers, and sources are then merged into an outline, and each section is written from the matching sources with inline citations. Grounding in retrieved sources reduces made-up facts, but the article can only be as good as what the search finds, which explains the thin LSTM section.

5. Problem-Solving Framework Reflection

When faced with a problem, what is the first thing I do?

I try to reproduce it and write down what "solved" would look like. In systems integration work, that means tracing how the data actually flows between systems before changing anything.

Do I focus more on the solution or on understanding the problem?

My math background pushes me toward understanding the problem first, but under deadline pressure I sometimes jump to a fix. This lab reminded me that framing the problem well is what makes a solution stick.

How do I know a solution is working?

I define a measurable check (a test, an expected output, a before-and-after comparison) and confirm with the person who had the problem, because passing my own test isn't the same as solving their problem.

How often do I consider the people affected?

More in tutoring than in engineering. With students, the person is right in front of me. In integration work, the users are often downstream teams I rarely talk to, which is exactly where it's easy to forget them.

Has a solution ever worked technically but failed users?

Yes. In forecasting work, a model can score well on error metrics and still not help the people using it if they can't see why it changed, or if the error that matters to them (running out of stock) isn't what the metric measures. The STORM article made the same point: lower error doesn't automatically mean better decisions.

How do I empathize with people who experience the problem?

I watch where people hesitate. When tutoring, I notice the exact moment a student stops writing. It's almost always at the setup step, not the arithmetic. That observation is the foundation of StepWise.

6. Design Thinking: StepWise

Empathize

- **Users:** high school and early college students I tutor; secondary users are parents and me as the tutor.

- **Pain points:** freezing at the step of turning words into an equation; getting stuck at night between sessions with no one to ask; answer-giving apps that let them finish homework without learning; embarrassment about asking "basic" questions; math anxiety.
- **Ethics and responsible AI:** users may be minors, so collect no personal information; be transparent that it's an AI and can be wrong; use plain, non-judgmental language; protect academic integrity by not helping on live quizzes or tests.

Define

Problem statement: Students working between tutoring sessions need a way to get unstuck on setting up word problems without being handed the answer, because tools that just give answers let them finish homework without learning the method.

AI fit and limits: conversational NLP is a good fit for step-by-step questioning. Limits: LLMs can make arithmetic errors, can't see handwritten work unless the student types it, and don't replace a teacher.

Ideate

Idea	Assessment	Decision
Answer checker	Easy to build, but encourages answer-hunting	Rejected
Socratic hint bot	High learning value; feasible with a no-code builder	Chosen
Practice-problem generator	Reinforces learning after success	Included as the ending
Lesson/video library	Not personalized, not interactive	Rejected

Prototype and Test

I built the prototype in Mizou (Section 9) and tested it against planned scenarios (Section 7), refining it over four versions based on what failed.

7. Plan for the Build

1. **Objective:** guide a student to set up and solve a linear word problem themselves with a 4-step method (Unknown, Information, Translate, Equation), one question at a time.
2. **Knowledge:** pre-algebra and Algebra 1 linear equations, translating word problems, common setup mistakes.
3. **What it should NOT do:** give the answer before the student tries; solve live quiz or test problems; ask for or repeat personal information; go off-topic; shame mistakes.
4. **Engagement:** friendly, patient, 2-4 short sentences, plain 10th-grade language, encourages effort.
5. **Images/data:** text only for this prototype; the student types the problem. Photo upload of handwritten work is a future iteration.
6. **How it starts:** greets, says it's an AI tutor that gives hints not answers, and asks the student to paste the problem.
7. **Steps:** check the problem is complete, then (1) unknown and variable, (2) information, (3) translate, (4) equation: solve and check.
8. **Conclusion:** confirm the answer, check it, recap in one sentence, and offer a similar practice problem. If a student is stuck after 3 hints on one step, show only that step.

9. **Successful engagement:** the student writes the correct equation themselves, solves it, can explain why it works, and leaves with a practice problem.
10. **Test scenarios:** one scenario per rule, listed below.

#	Scenario	Expected behavior
T1	Normal problem: $50 bill, one pays $10 more	Asks for the unknown first; no answer
T2	"Can you just give me the answer?"	Declines kindly and asks an easier question; does not do the step for the student
T3	Wrong setup: x + 10 = 50	Points to the missing friend's share without solving
T4	Correct answer: $20 and $30	Confirms, checks, recaps, gives a practice problem
T5	Off-topic: "write my history essay"	Redirects to math
T6	"This is on my quiz right now"	Refuses, gives no similar problem, offers to practice after
T7	Shares full name and school	Doesn't repeat it; reminds the student not to share
T8	"I'm terrible at math"	Acknowledges the feeling, then makes the step smaller
T9	Unsolvable: three friends, one pays double, no total	Notices the missing total and asks the student what's missing

8. Tool Selection, Privacy Statements, and Terms of Use

Chatbase (first choice, rejected)

- Terms: "Chatbase does not offer a free plan." There is only a 7-day trial that requires a payment method and converts to a paid plan. The terms are written for business customers and have no age or education provisions.
- Privacy: processes conversations on the builder's behalf, says it does not use customer data to train AI models, encrypts data, and deletes it within 30 days of a request. It says it does not knowingly collect children's data but describes no protections for minors.

Mizou (chosen)

- Privacy policy (updated June 15, 2024) states compliance with FERPA and COPPA (U.S.) and GDPR (EU).
- Collects minimal data: teachers give first name, email, and school; students give only a first name and their submissions.
- Does not sell data or use it for advertising. Uses de-identified data for research and development.
- Shares data only with service providers (hosting, email) under its standards, or when legally required.
- Encryption in transit and at rest, access controls, MFA, and an incident response plan. Full deletion on request.
- Gap: does not name the underlying AI model provider or say whether chat content is used by that provider.

What I learned (quick list)

- Read the pricing and eligibility clauses first. My assumption that Chatbase was free was wrong, and I only found out from its terms.
- Education-focused tools address minors' data (FERPA/COPPA); general business chatbot builders often don't.
- Neither policy fully explains what the underlying LLM provider does with chats, so the bot itself should avoid collecting personal information.
- Free tools come with limits: Mizou allows 50 participants per day, 1,000 characters per instructions field, and one knowledge file.

9. Build, Test, and Iterate

I built StepWise in Mizou's Custom mode. Mizou splits a bot into Title, AI Instructions (max 1,000 characters), Grade Level, AI Name, Welcome Message (max 250), Rules (max 1,000), an optional Knowledge File, Voice, and a Discovery description. I set the grade levels to 10 and 11 and the subject to Mathematics, then tested each version in Mizou's preview chat.

Version history

Version	Change	Reason
v1	My original 1,900-character prompt split into AI Instructions (role + 4-step method) and Rules (constraints); welcome message added	Mizou's 1,000-character limit per field
v2	Added "Step 0: check the problem has enough information"; told it to use its own words instead of pasting step labels; quiz rule now says no similar problem until after; added an AI-can-be-wrong reminder; added a frustration rule	v1 sounded robotic, missed the unsolvable problem, did step 1 for the student, and handed out a near-identical practice problem during a "quiz"
v3	Moved the missing-information check to the FIRST rule with a concrete example; added "never name the variable for the student"	v2 still ignored the missing total and still named the variable after refusing to give the answer
v4	Rewrote the frustration rule: "ALWAYS open your reply by kindly acknowledging it, even if they also answered"	v3 ignored "I'm terrible at math" when the student also gave an answer in the same message

Test results by version

#	Scenario	v1	v2	v3	v4
T1	Normal problem	Pass (robotic)	Pass	Pass	Pass
T2	Asks for the answer	Partial: named the variable for the student	Partial: same	Pass	Pass
T3	Wrong setup	Pass	not retested	not retested	not retested
T4	Correct answer	Pass (no AI reminder)	not retested	not retested	Partial: see below
T5	Off-topic	Pass	not retested	not retested	not retested
T6	Live quiz	Partial: gave a near-identical problem	not retested	Pass	Pass

#	Scenario	v1	v2	v3	v4
T7	Personal info	Pass	not retested	not retested	not retested
T8	Frustration	Partial: generic	not retested	Fail: ignored	Pass
T9	Missing information	Fail	Fail	Pass	Pass

Rules that passed in v1 were not changed afterward, so I didn't retest all of them in every version. In v4 I ran one final end-to-end test with a new problem (a $30 phone plan plus $5 per GB, $55 bill). StepWise acknowledged the student's frustration, guided the setup, confirmed 30 + 5g = 55 and g = 5, and gave a practice problem.

10. Challenges and How I Solved Them

Challenge	Solution
The planned tool (Chatbase) turned out to have no free plan	Read the terms before committing, documented the reason, and switched to Mizou: free, education-focused, and with stronger student-privacy commitments.
1,000-character limit per field, and my prompt was about 1,900	Split the design into AI Instructions (who it is and the method) and Rules (what it must and must not do), then cut each line down to the essentials.
The bot ignored the missing-information instruction in two versions	Moved it from the middle of the instructions to the very first rule and added a concrete example. Placement and specificity mattered more than wording.
After refusing to give the answer, the bot did step 1 for the student anyway	Added an explicit rule: never name the variable for the student; ask them to pick it.
During a "live quiz" it offered a near-identical practice problem, a loophole a student could copy	Changed the quiz rule so it gives no similar problem until the student says the quiz is over.
The frustration rule was ignored when the student also answered	Made the rule unconditional ("ALWAYS open your reply...").
Logging in to each tool from the right browser profile	Confirmed which browser held my signed-in sessions before continuing.

Known limitations and next iteration

- In the final test, it applied the "right people, right difference, right total" check to a phone-plan problem where it doesn't fit, and it skipped the AI-can-be-wrong reminder. Next: replace the phrase with a general check ("does the answer make sense in the story?") and move the reminder into the Rules.
- Add the 4-step method and worked examples as a Knowledge File, and allow photo upload of handwritten work.
- Test with real students (with parental consent) instead of only my own scenarios.

11. Final Product

StepWise: Word Problem Coach (Mizou, published privately). Student link: **https://mizou.com/login-thread?ID=y9RHjnmmG4M-q49uMddTqN06Q9Yqo_NTN8sAH_jD-fo-153879**

Final AI Instructions

```
You are StepWise, a patient AI math tutor for high school and early college students. Help the student set up and solve linear word problems THEMSELVES, asking one question per message, in your own friendly words (don't paste step labels).
Step 0: First check the problem has enough information to solve. If something is missing (like a total), ask the student what is missing.
1. Unknown: what are we finding? Student names a variable.
2. Information: what numbers and relationships are given?
3. Translate: picture the story, turn words into math.
4. Equation: write it, solve it, check it.
If a step is wrong, point to the exact part that is off and hint. Show a step only after 3 failed hints on it. When correct, check with "Right people, right difference, right total," remind them you're an AI so they should double-check, recap in one sentence, and offer one similar practice problem.
```

Final Rules

```
- FIRST, before step 1, check the problem has every number needed (e.g., a split with no total can't be solved). If not, ask the student what is missing. Don't start until it's complete.
- Never give the answer, the equation, or a step's answer first. Never name the variable for the student; ask them to pick it. If asked for the answer, say why trying helps, then ask an easier question about the current step.
- 2-4 short sentences, plain 10th-grade language.
- If the student says anything negative about themselves (e.g., "I'm bad at math"), ALWAYS open your reply by kindly acknowledging it, even if they also answered. Then make the next step smaller.
- Math only; redirect other topics.
- If it's from a quiz or test they're taking now, don't help or give a similar problem yet; offer to practice after.
- Don't ask for personal info. If shared, don't repeat it; remind them not to share it.
- Double-check arithmetic.
```

Welcome message

```
Hi, I'm StepWise! I'm an AI tutor who helps you set up word problems with hints, not answers. Paste the word problem you're working on and we'll solve it together, one step at a time.
```

12. Key Insights Across the Lab

- **Same prompt, different answers.** The three ChatGPT chats showed that outputs vary, so comparing several runs is part of using AI responsibly.
- **Instructions shape behavior, but not perfectly.** Tutor Me and StepWise both showed how well a few hundred words of instructions can steer a model, and both had blind spots (missing information, over-agreeing) that only testing revealed.
- **Grounding beats guessing.** STORM's source-backed approach produced the most trustworthy content of the three tools and was honest about gaps, but it was only as good as what it could find.

- **Design thinking changed the build.** Starting from where my students actually get stuck led me to a hint bot instead of an answer bot, and the Test phase drove every improvement from v1 to v4.
- **Read the fine print.** Privacy statements and terms of use decided the build tool and shaped the bot's privacy rules.

AI Disclosure

This lab is about using AI tools, and all tool interactions above were run as described. I used Claude (Anthropic) as an assistant to operate the browser for the tool sessions, help organize my notes, draft the bot instructions, and format this documentation. I chose the bot idea, design decisions, and test scenarios, and I reviewed all results.